

Architecture Selection and Justification

Remote Patient Vitals Alert and Monitoring App

I selected Microservices Architecture for secure remote patient monitoring. My requirements cover authenticated telemetry, validation and duplicate prevention, connection loss, critical-alert escalation and an authorized caregiver dashboard. The seven components in the diagram separate these responsibilities while retaining a common patient identity.

Architectural style comparison

Style	Assessment for this system
Layered	Simple separation of presentation, domain logic and data access. A conventional single deployment makes telemetry and dashboard releases and scaling interdependent.
Client-Server	Naturally connects remote devices and caregivers to a backend, but alone does not specify how validation, connection monitoring and alert processing are isolated.
Microservices	Allows independent telemetry and alert services with targeted scaling and fault isolation. Requires extra monitoring, network calls and careful state consistency.

Two reasons for this choice

1. Protect the telemetry path. The Ingestion Service validates fields and reading IDs, while the Analysis Service evaluates accepted readings. A correctly formatted critical value is preserved for evaluation. These services can scale separately from dashboard queries as the system approaches 500 simultaneous streams (FR-001, FR-002 and NFR-001).
2. Keep monitoring during delivery failures. Pending alerts are persisted before notification workers attempt delivery. Bounded timeouts and retries isolate a slow provider from ingestion. Ingestion separately records disconnection and recovery, allowing the dashboard to label old readings stale. Escalation stops after acknowledgement (FR-003 to FR-005).

Security advantage

Device credentials are checked against patient pairing; caregiver tokens are checked against assigned patients at each data API. TLS 1.2 or later protects transport, and AES-256 protects stored patient data. Restricted service credentials and audit records for access, configuration and acknowledgements support NFR-002. Separate service boundaries limit unnecessary data access.

Performance benefit and availability

Replicated analysis workers and versioned threshold caching target detection within 2 seconds of receipt (FR-004). Shared connection state avoids false disconnections when ingestion workers change. Supporting 500 concurrent streams and 99.99% monthly availability requires load tests, redundant instances and monitored failover. The Patient Data Service needs partitioning and replicas to avoid a bottleneck. These are proposed design targets, not benchmark results.